

Senior Full Stack Developer

Vijaya Krishna Nunna

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Professional Summary:

- Over **10 +** years of experience as Senior Full Stack .NET Developer delivering robust and scalable applications using **.NET Core, C#, ASP.NET Core, and Node.js** across various industries, including finance, healthcare, and government.
- Proficient in building **Microservices Architecture** for highly scalable, secure, and fault-tolerant systems, leveraging **RESTful APIs, GraphQL, and Event-Driven Architecture** with **Kafka** and **RabbitMQ**.
- Extensive experience in frontend development using **Angular (up to Angular 20), React, HTML5, CSS3, TypeScript, and JavaScript**, ensuring responsive and interactive UI/UX with frameworks like **RxJS** and **Angular Material**.
- Expertise in designing and managing databases using **SQL Server, MongoDB, PostgreSQL, Azure Cosmos DB, and Redis**, including writing optimized **T-SQL** stored procedures and managing ETL pipelines with **SSIS**.
- Skilled in implementing secure authentication and authorization using **OAuth2, OpenID Connect, JWT**, and integrating with **Azure AD** and **ASP.NET Identity** for user access management.
- Advanced knowledge of cloud platforms like **Azure** and **AWS**, with hands-on experience in **Azure App Services, Azure Blob Storage, Azure Key Vault, Azure API Management, AWS Lambda, AWS S3, and Azure DevOps** for CI/CD pipelines.
- Proficient in containerization using **Docker** and orchestration with **Kubernetes**, enabling scalable, consistent deployments across environments.
- Adept at integrating real-time communication and peer-to-peer streaming using **SignalR, WebRTC**, and ensuring data flow efficiency with **RxJS** in frontend applications.
- Demonstrated expertise in backend frameworks, implementing **Dependency Injection** for maintainable and testable services, and following architectural patterns like **Onion Architecture, MVC, and Service-Oriented Architecture (SOA)**.
- Strong focus on code quality with tools like **SonarQube**, thorough testing with **XUnit, Karma, Jasmine**, and API validation using **Postman** and **Swagger/OpenAPI**.
- Hands-on experience in managing and optimizing application performance, including caching strategies with **Redis, Azure Cache for Redis, and MongoDB Atlas** for NoSQL databases.
- Designed and delivered enterprise-level reports and dashboards using **SSRS** and advanced visualizations in **Angular**, enabling data-driven decision-making.
- Specialized in integrating compliance standards like **HIPAA, FHIR, and HL7** for healthcare systems and implementing **SSL/TLS** protocols for secure data transmission.
- Skilled in leveraging centralized logging and monitoring solutions like **Log4Net** and troubleshooting distributed applications for error resolution and performance optimization.
- Experienced in iterative development using **Agile SDLC methodologies**, managing tasks with **JIRA**, and ensuring smooth team collaboration with **Git** and **Azure Boards**.
- Strong knowledge of serverless architectures using **Azure Functions** and **AWS Lambda**, delivering cost-efficient and scalable solutions for event-driven applications.

Technical Skills:

Category	Technologies
Languages	C, C#, .NET, Web API, ASP.NET, MVC, TypeScript, VB, Python, MVC4, JavaScript, WinForms, VBScript, XHTML5
Databases	SQL Server, Entity Framework Core, SQL Server 2012/2008/2005, PL/SQL, T-SQL, Oracle, IBM DB2, Microsoft Access
Web Technologies	ASP.NET Core, Angular, HTML5, CSS3, JavaScript, React, CSS (Bootstrap), Appian
Data Access Technologies	ADO.NET, ADO, DAO, OLE DB, ODBC, LINQ, Hibernate
.NET Technologies	Web Services, Web/Win Forms, Windows Services, LINQ, WCF, Entity Framework
Webserver	.NET Core, .NET, ASP.NET Core, Angular, .NET
Framework	IIS, ASP.NET Web Development Server
Application Tools	Visual Studio, MS Office, TOAD, Azure DevOps, Git, JIRA
Operating Systems	Windows 11, XP/NT/2000/9x, MS-DOS, UNIX
Cloud Technologies	Azure (Azure DF, Azure Blob Storage, Azure DevOps), AWS

Professional Experience:

Commonwealth Of Massachusetts | Boston, MA

Jan 2025 to Present

Sr. DotNet Developer

Responsibilities:

- Drove **Agile/SCRUM** delivery through sprint planning, daily stand-ups, sprint reviews, and retrospectives, tracking work items in **Azure Boards** and coordinating with cross-functional teams to consistently ship high-quality, on-time releases.
- Architected enterprise-grade applications on **.NET 10, ASP.NET Core Web API, and C#**, applying **Clean Architecture** and **Microservices Architecture** principles to maximize scalability and long-term maintainability.
- Implemented **GitHub Copilot** and **Azure AI-assisted code review** across the development team, accelerating feature delivery on Appian and .NET modules while maintaining secure coding standards for government systems.

- Designed an **agentic AI workflow** using **Azure OpenAI** and **Model Context Protocol (MCP)**-based tool integration to automate multi-step case processing within Appian, enabling the agent to retrieve records, apply business rules, and route exceptions for human review.
- Engineered responsive, accessible front-end experiences using **Angular 20**, **TypeScript**, **RxJS**, **HTML5**, **CSS3**, and **Bootstrap**, modernizing legacy state agency interfaces for improved usability.
- Developed select **React**-based modules and reusable components for newer state agency dashboards, integrating them with existing **Angular 20 applications during an incremental frontend modernization effort**.
- Built and maintained low-code **Appian 24.4x/25.x** applications leveraging **SAIL**, **Records**, **Process Models**, **Expression Rules**, **Decision Rules**, and **Integration Objects** to automate high-volume government business processes.
- Unified **Appian** with **ASP.NET Core Web APIs** and external state systems through **REST API** integrations, establishing seamless, real-time data exchange across multiple agency platforms.
- Engineered reusable **Connected Systems**, **Web APIs**, and **Integration Objects** within **Appian** to consume secure internal and third-party services, cutting redundant integration effort across teams.
- Elevated workflow transparency and operational reporting by configuring **Appian Records**, **Record Actions**, **Record Types**, and **Process Reports** for real-time process insight.
- Fortified data protection by configuring **Appian Sites**, security groups, role-based access controls, and object-level permissions safeguarding sensitive government information.
- Enforced enterprise-grade identity security using **Azure AD**, **OAuth 2.0**, **OpenID Connect**, and **JWT**, securing authentication and authorization across all application layers.
- Tuned **SQL Server** performance via custom stored procedures, functions, views, and triggers, while streamlining data access with **Entity Framework Core** and **LINQ**.
- Enabled resilient, decoupled communication between **Appian** workflows and **.NET microservices** using **Azure Service Bus**, and automated scheduled processing with **Azure Functions**.
- Containerized and orchestrated **.NET microservices** with **Docker** and **Azure Kubernetes Service (AKS)**, backed by **Azure App Services**, **Azure Blob Storage**, and **Azure Key Vault** for secure hosting and secrets management.
- Standardized release management by building and maintaining **Azure DevOps CI/CD** pipelines spanning both **.NET** and **Appian** deployments, automating build validation and testing.
- Safeguarded application quality through **xUnit** unit testing, **Jasmine/Karma** front-end testing, and API validation with **Postman** and **Swagger/OpenAPI**.
- Established full-stack observability with **Serilog**, **Azure Application Insights**, and **Azure Monitor**, accelerating production issue detection and resolution.
- Partnered with business analysts, QA engineers, **Appian** developers, and agency stakeholders to translate requirements into scalable solutions and resolve production issues.
- Upheld secure coding standards and government accessibility/compliance requirements through rigorous, ongoing code reviews.

Environment: .NET 10, C#, ASP.NET Core Web API, Entity Framework Core, Angular 20, React, TypeScript, HTML5, CSS3, Bootstrap, RxJS, Azure OpenAI, RAG, PCI DSS, Appian 24.x/25.x, SAIL, Records, Process Models, Integration Objects, Connected Systems, Web APIs, SQL Server 2022, Azure App Services, Azure Functions, Azure Service Bus, Azure Blob Storage, Azure Key Vault, Azure AD, OAuth 2.0, OpenID Connect, JWT, Redis Cache, Docker, Kubernetes (AKS), Azure DevOps, Git, xUnit, Jasmine, Karma, Postman, Swagger/OpenAPI, Serilog, Azure Application Insights, Azure Monitor, Visual Studio 2022, JIRA.

Brown Brothers Harriman | New York, NY

Senior Full Stack Developer

June 2023 to Jan 2025

Responsibilities:

- Utilized **Agile SDLC** methodologies for continuous delivery and iterative development, ensuring flexibility in adapting evolving business requirements and timely feature releases for the **Client Portal Management** application.
- Designed and implemented a **Microservices architecture** to ensure scalability, security, and fault tolerance, optimizing system performance and integration in the financial data management platform.
- Developed backend services using **C#** and **ASP.NET Core Web API**, building robust **RESTful services** to interact with financial data and ensuring maintainable and secure data exchange.
- Leveraged **ASP.NET Core Dependency Injection** to improve the maintainability of backend services by promoting loose coupling and enhancing testability of business logic.
- Developed and optimized **RESTful APIs** using **C#** and **ASP.NET Core**, implementing **asynchronous programming** with **async/await** to enhance performance and scalability, ensuring efficient handling of high-volume financial data requests.
- Utilized **Entity Framework Core** to interact with relational databases, simplifying data access and ensuring efficient querying of financial data in **SQL Server**.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline with **Azure OpenAI** and a vector database, integrated into a **human-in-the-loop AI agent** that answered client-portal account queries and escalated ambiguous requests, cutting first-response time for common questions.
- Evaluated LLM output quality using a lightweight **prompt evaluation framework**, tracking accuracy and hallucination rate for an internal AI assistant before production rollout, ensuring reliable responses for financial data queries.
- Developed **Python** scripts for automating data validation and reconciliation tasks across financial data pipelines, reducing manual processing time, and improving data accuracy for downstream reporting.
- Ensured secure handling of sensitive financial data in compliance with **PCI DSS** standards, implementing encryption, access controls, and audit logging across client-facing APIs and data storage layers.
- Developed visually engaging and functional web applications using **HTML**, **CSS**, and **JavaScript**, ensuring an intuitive user experience for financial data processing.
- Implemented real-time data features in **Angular 16** using **RxJS** to manage data flow efficiently, improving user engagement by consuming backend API data in the financial application.

- Utilized **Angular CLI** for efficient project scaffolding, build optimization, lazy loading, and **Angular Universal** for server-side rendering (SSR), enhancing SEO, initial load performance, and secure communication with **JWT** for authenticated data exchange in financial applications.
- Enhanced **UI/UX** by integrating **Angular Material**, creating consistent, accessible user interfaces that ensured a seamless experience across platforms for financial application users.
- Integrated **JSON** as the primary data format for communication between backend services and the frontend, enabling lightweight data transfer and efficient data handling for financial transactions.
- Secured APIs with **OAuth2** for user authentication and authorization, ensuring only authorized users could access sensitive financial data and transactions.
- Worked with **SQL Server** to manage financial records, ensuring efficient storage, retrieval, and modification of financial data while maintaining data integrity and security.
- Designed and optimized **T-SQL** stored procedures and triggers for efficient querying and manipulation of financial data within **SQL Server**.
- Integrated **MongoDB** for NoSQL storage, ensuring flexible and scalable data management for semi-structured financial data in a microservices architecture.
- Integrated **Apache Kafka** for event-driven communication between distributed microservices, ensuring real-time processing of financial data and reducing latency.
- Utilized **Postman** and **Swagger UI** for comprehensive **API testing**, ensuring all endpoints adhered to defined standards and were functioning correctly.
- Leveraged **Azure Cloud** for hosting and deploying applications, using **API Gateway** for secure routing of financial API requests and ensuring high availability and reliability of services.
- Worked with **Azure Blob Storage**, **Azure Key Vault**, and **Azure App Service** for storage management, secret handling, and application hosting in the cloud.
- Built and maintained **CI/CD pipelines** using **Azure DevOps**, automating build, testing, and deployment processes to streamline code delivery and application deployment.
- Containerized applications using **Docker** and orchestrated deployments with **Kubernetes**, ensuring scalability and efficient deployment across multiple environments.
- Implemented **GraphQL** endpoints alongside RESTful APIs to give front-end teams flexible, efficient querying of financial data, reducing over-fetching and improving Angular application load times.
- Conducted testing for backend services and front-end components using **XUnit**, **Karma**, and **Jasmine**, ensuring functionality, reliability, and cross-browser compatibility in the financial data management system.
- Implemented centralized **logging** using **Log4Net** to monitor errors and events across various services, enabling quick issue resolution and operational efficiency.
- Tracked project progress using **JIRA** for task management and timelines, while utilizing **Git** for branching and merging strategies to ensure seamless team collaboration and efficient handling of development features and bug fixes in the financial data management system.

Environment: C#, ASP.NET Core, Angular 16, TypeScript, SQL Server, MongoDB, Python, Redis, Kafka, Docker, Kubernetes, Azure App Services, OAuth2, JWT, Git, xUnit, Postman, Swagger, SonarQube, GraphQL, Log4Net, JIRA, Azure Functions, Entity Framework Core.

UnitedHealth Group | Plymouth, MN

Dec 2020 to May 2023

.Net Developer

Responsibilities:

- **Agile** approaches were used, emphasizing iterative development, frequent sprint planning, and ongoing feedback to improve teamwork and project flexibility.
- Implemented **Onion** architecture to separate business logic, user interface, and user input handling, enhancing maintainability and scalability of the applications.
- Worked on the **Healthcare Data Management & Compliance Systems** application, building dynamic and interactive user interfaces with **Angular 14**, improving user experience and ensuring smooth integration with backend services.
- Developed and optimized enterprise-level **RESTful APIs** with **C#**, ensuring secure, efficient system integration with other Healthcare platforms and services.
- Delivered high-quality web services using **ASP.NET Core**, focusing on modular design, security, and performance to meet the needs of healthcare data management.
- Utilized **Entity Framework Core** for object-relational mapping with **SQL Server**, simplifying data access and manipulation.
- Built dynamic dashboards with **Angular 14**, visualizing real-time healthcare data such as patient information, compliance metrics, and performance indicators, enabling healthcare providers to make informed decisions.
- Managed real-time data flow by utilizing **Angular services** and **RxJS**, optimizing asynchronous operations and boosting **Healthcare Data Management** application performance.
- Integrated secure patient login systems with **Angular 14** using **OAuth 2.0** and **OpenID Connect**, ensuring HIPAA-compliant authentication and authorization for users accessing sensitive health information.
- Secured API communications using **HTTPS/TLS** and **JWT**, managing **SSL certificates** to ensure safe and reliable data exchanges.
- Designed and managed databases using **SQL Server**, **MongoDB**, and **Azure Cosmos DB**, implementing **stored procedures**, **triggers**, and **functions** to ensure data integrity and optimize performance.
- Enhanced backend workflows with **T-SQL**, automating business processes to improve overall system efficiency.
- Integrated **Redis** for caching, significantly improving the performance of frequently accessed data.
- Developed and maintained robust **RESTful APIs**, handling complex data queries and system integration for real-time healthcare data sharing.
- Implemented **SignalR** for real-time communication and **WebRTC** for peer-to-peer streaming, enhancing the responsiveness and interactivity of the application.

- Integrated **FHIR standards** and **HL7 message parsing** to enable seamless healthcare data exchange, ensuring compliance with healthcare industry regulations.
- Ensured **HIPAA compliance** by implementing encryption and secure data transmission in all healthcare-related processes.
- Leveraged **Kafka** for message brokering between microservices, improving scalability and real-time data processing.
- Utilized **RabbitMQ** for reliable asynchronous messaging between healthcare microservices, ensuring guaranteed delivery of compliance and patient data events across distributed systems.
- Utilized **Docker** and **Kubernetes** for containerization and orchestration, ensuring consistent environments and scalability.
- Managed user identities with **Azure AD** and stored unstructured data in **Azure Blob Storage** for better security and management.
- Implemented serverless architecture with **Azure Functions** and secured sensitive data using **Azure Key Vault**.
- Automated **CI/CD pipelines** using **Azure DevOps**, speeding up development cycles and ensuring consistent deployment across environments.
- Designed a **Service-Oriented Architecture** using **WCF**, optimizing service communication between various systems.
- Managed project workflows with **JIRA**, ensuring timely issue resolution and development progress.
- Conducted comprehensive testing using **Jasmine**, **Postman**, and **Selenium**, ensuring reliable and efficient functionality across the system.
- Managed source control using **Git**, supporting version control and seamless collaboration among distributed teams working on financial software.

Environment: .NET Core 6, C#, Angular 14, TypeScript, Entity Framework Core, SQL Server, MongoDB, Redis, FHIR, HL7, RESTful APIs, SignalR, OAuth 2.0, JWT, Docker, Kubernetes, Azure App Services, Azure Functions, Azure DevOps, Azure AD, JIRA, CI/CD, Jasmine, Kafka, RabbitMQ, WCF, HIPAA compliance.

Ingram Micro | Irvine, CA

Aug 2018 to Nov 2020

.Net Web Developer

Responsibilities:

- Designed and deployed **microservices** to modularize billing functions, enabling scalable and independent management of services and improving system resilience.
- Implemented **MVC architecture** to structure the application for scalable and maintainable code, enhancing user experience and separation of concerns.
- Styled and structured web pages using **HTML5** and **CSS3**, ensuring responsive design, modern aesthetics, and a seamless user experience across different devices and browsers.
- Developed client-side functionality using **TypeScript** for interactive and real-time features.
- Developed dynamic and responsive web applications using **Angular**, utilizing its powerful features like data binding, component architecture, and dependency injection to deliver high-performance user interfaces and seamless functionality.
- Leveraged **C#** and maintained backend services that supported business processes and customer-facing apps by integrating **APIs**, data processing, and essential business logic.
- Built **RESTful APIs** with **ASP.NET Core** and **ASP.NET Web API**, enabling secure and efficient communication between client and server.
- Integrated **OAuth/OpenID Connect** and **ASP.NET Identity** for secure user authentication and authorization.
- Designed and implemented **PostgreSQL databases** to efficiently store, manage, and retrieve large datasets, ensuring high availability and data integrity while optimizing query performance.
- Designed and enhanced data access layers using **ADO.NET** with **PostgreSQL**, improving query performance and efficiently managing database connections.
- Performed unit testing with **xUnit** to ensure code reliability and automated UI testing with **Selenium** for frontend validation.
- Verified **API** functionality using **Postman** and documented APIs with **Swagger/OpenAPI** for interactive and clear documentation.
- Managed and secured APIs through **Azure API Management**, ensuring proper access control and monitoring.
- Leveraged **Azure AD** for centralized user identity and access management.
- Managed large datasets with **Azure Blob Storage** and protected application secrets and sensitive data using **Azure Key Vault**.
- Utilized **Azure Cosmos DB** for flexible, scalable NoSQL data management where needed.
- Managed application lifecycle with **Azure DevOps**, including **YAML Pipelines** for CI/CD and version control using **GitLab CLI**.
- Leveraged **AWS Lambda** for serverless event processing and **AWS S3** for scalable storage of billing documents and reports, reducing infrastructure overhead for high-volume batch jobs.
- Leveraged **Docker** for containerizing applications to ensure consistent deployment across environments, while utilizing **Kubernetes** for orchestration, automating scaling, monitoring, and maintaining high availability for production systems.
- Designed and implemented detailed billing reports using **SQL Server Reporting Services (SSRS)**, ensuring accurate financial tracking and providing actionable insights for decision-making.
- Utilized **SQL Server Integration Services (SSIS)** to automate and optimize ETL processes, improving data quality and streamlining the flow of information for reporting and analysis.
- Utilized **WCF security features** like message encryption, authentication, and authorization, ensuring secure communication for sensitive business data.

- Implemented data binding, commands, and validation in **WPF** to ensure smooth interaction between the user interface and the underlying business logic, enhancing application responsiveness and usability.

Environment: .NET, C#, ASP.NET Core, ASP.NET Web API, ADO.NET, PostgreSQL, SQL Server, Azure Cosmos DB, MVC architecture, Angular, HTML, CSS, TypeScript, OAuth/OpenID Connect, ASP.NET Identity, Azure AD, HTTPS/TLS, DICOM, SSRS, HIPAA, Event Hub, Azure DevOps, YAML Pipelines, GitLab CLI, Docker, Kubernetes, Azure Blob Storage, Azure Key Vault, AWS Lambda, AWS S3, xUnit, Selenium, Postman, Swagger/Open API, Azure API Management, Azure Service Bus.

CSS Corp Pvt Ltd | Chennai, India **May 2016 to Aug 2018**

.NET/J2EE Developer

Responsibilities:

- Implemented **Waterfall SDLC methodology**, ensuring structured, phased project management from requirements gathering through to final delivery, adhering to timelines and stakeholder expectations.
- Designed responsive and intuitive front-end interfaces using **HTML, CSS, JavaScript, jQuery, Ajax, and WebForms**, optimizing the user experience across devices.
- Developed and optimized web applications using the **.NET Framework**, including **WebForms**, driving improvements in both front-end user experience and back-end functionality.
- Designed and deployed **SOAP Web Services**, facilitating secure, efficient communication between distributed systems and enhancing integration capabilities across platforms.
- Utilized **ADO.NET** for seamless database access, improving performance and ensuring reliable interactions with **SQL-based** data sources.
- Managed databases with **SQL Server, Oracle, and PostgreSQL**, ensuring optimal performance, data integrity, and secure handling of business-critical data.
- Developed and optimized complex **PL/SQL queries** to efficiently retrieve, manipulate, and store data in **Oracle** databases, ensuring high performance and reliability.
- Developed high-performance, secure web services using **WCF**, enabling seamless communication between distributed applications with varying protocols like **HTTP, TCP, and MSMQ**.
- Integrated **WPF applications** with backend services such as **REST APIs** and **WCF**, enabling real-time data updates, and providing end-users with accurate, up-to-date information.
- Conducted thorough unit testing using **xUnit** and **MSTest**, ensuring functionality and reliability of new features prior to deployment.
- Applied **SVN** for version control, ensuring efficient code management, collaboration, and change tracking across development teams.

Environment: .NET Framework, C#, WebForms, HTML, CSS, JavaScript, jQuery, Ajax, SOAP Web Services, ADO.NET, SQL Server, Oracle, PostgreSQL, PL/SQL, WCF, HTTP, TCP, MSMQ, WPF, REST APIs, xUnit, MSTest, SVN

Education:

Bachelor's in computer science & engineering | NEC, AP, India